

Climate 5000 Series - Gen 4

48K BTU Single Zone Regular Light Commercial System

4-Way Cassette IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

48K Btu Single Zone Regular Light Commercial System - 4-Way Cassette IDU

Indoor part number 8-733-962-683

Indoor model number BMS500-AAU048-1AHCXD

Outdoor part number 8-733-962-693

Outdoor model number BMS500-AAS048-1CSXLD

Efficiency Data AHRI 210/240 2023

SEER2 19

EER2 19

HSPF2 Region IV 9.9

HSPF2 Region V 7.7

COP(47F) 2.96

COP (5F) 2.11

Performance Data AHRI 210/240 2023

Cooling	Capacity Range	10600~50400
	Rated Capacity	48000
	Current	22.5
	Input Power	5053

Heating	Capacity Range	19900~61700
	Rated Capacity (47F)	50000
	Rated Capacity (5F)	40500
	Current	22.1
	Input Power	4950

Electrical Data

Power supply (V / Ø / Hz) 208-230 / 1 / 60

Min. circuit ampacity indoor/outdoor (A) 3.0 / 39.0

Max. fuse indoor/outdoor (A) 15.0 / 40.0

Compressor Data

Type: Rotary

Refrigerant oil /oil charge (ml) VG74 / 1400

Input (W) 3700

Rated current (RLA/A) 7.0

Piping

Refrigerant precharge (oz/kg) 126.99 / 3.60

Length for precharge (ft/m) 24.6 / 7.5

Liquid side (in/mm) 3/8" / Ø9.52

Gas side (in/mm) 3/4" / Ø19.0

Additional Charge of Refrigerant (oz/ft)/(g/m) 0.32 / 30

Max. refrigerant pipe length (ft/m) 246.00 / 75

Max. difference in height (ft/m) 98.40 / 30

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type R454B

Design pressure (PSIG) 550 / 340

Dehumidification (pts/hr) 13.25

IDU	Fan motor model	ZKFN-125-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	200.0
	Fan speed (RPM)	720 / 660 / 580
	Air flow, Turbo / H / M / L / Si (CFM)	1236.1 / 1124.2 / 1018.3 / 800.5 / 688.7
	Approx. Air Throw Range ft (m)	13.1 / 4
	Noise level, Turbo / H / M / L / Si (dB(A))	56.0 / 52.5 / 50.0 / 42.5 / 38.50
	Net weight (lbs / kg)	64.59 / 29.3
	Gross weight (lbs / kg)	76.06 / 34.5
	Panel net weight (lbs / kg)	13.23 / 6
ODU	Gross panel weight (lbs / kg)	19.84 / 9
	Drain pipe diameter (mm)	ODØ25mm
	Fan motor model	ZKFN-85-10-1L
	Fan motor qty (ea)	2
	Fan motor input (W)	126.0
	Fan motor speed (RPM)	850 / 750 / 650
	Air flow, Max (CFM)	4237.9
	Noise level, Max (dB(A))	62.5
	Net weight (lbs / kg)	208.78 / 94.7
	Gross weight (lbs / kg)	239.86 / 108.8

Accessories

BMS-RCG-XXD (8733962703) Decorative Grille - 24K ~ 48K

BMS-WT2-XXD (8733962707) Wired Wall Thermostat

BMS-DRY-XXD (8733969263) 24V Interface

Standard Parts

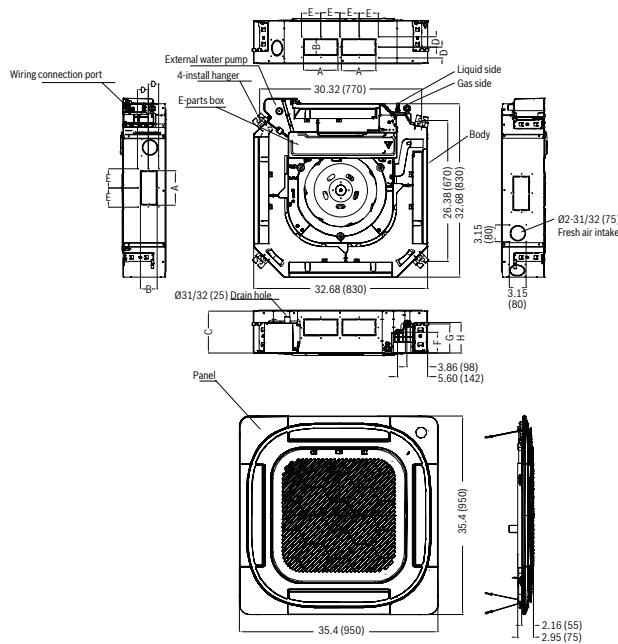
8-733-969-065 Remote Control

Climate 5000 Series - Gen 4

48K Btu Single Zone Regular Light Commercial System - 4-Way Cassette IDU

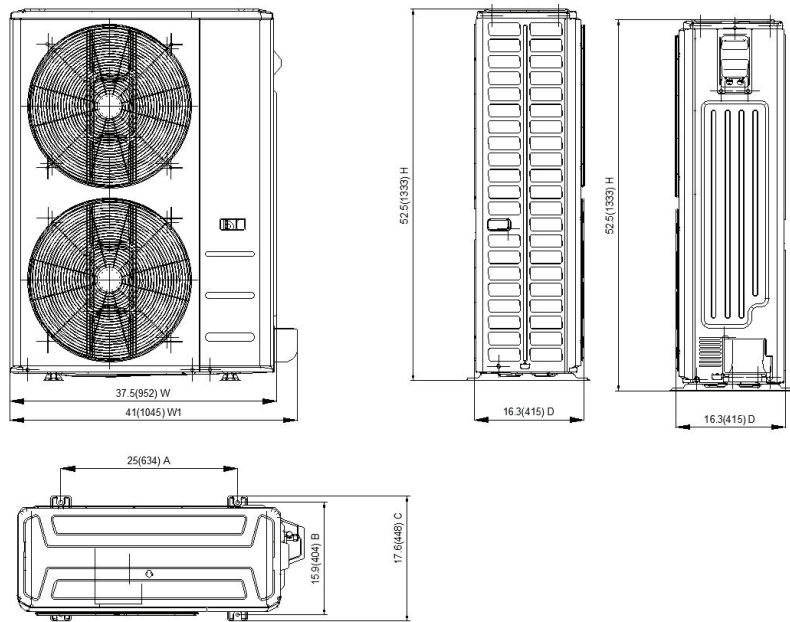


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU048-1AHCXD	32.68	32.68	11.30

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	A (in)	B (in)	C (in)
BMS500-AAS048-1CSXLD	37.48	16.34	52.48	41.0	25.0	15.9	17.6

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Cooling Performance - Indoor Dry Bulb Temperature at 1235 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F (16.0°C)					64.4°F (18.0°C)					67°F (19.4°C)					71.6°F (22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
	S/T	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
	PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
-13°F (-25.0°C)	TC	46168	46230	46280	46282	46344	48950	49008	49057	49066	49125	51875	51942	52000	52010	52068	55000	55060	55120	55129	55197
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	2.32	2.32	2.32	2.32	2.33	2.46	2.46	2.46	2.46	2.47	2.60	2.61	2.61	2.61	2.61	2.76	2.76	2.77	2.77	2.77
-4°F (-20.0°C)	TC	49719	49786	49840	49842	49909	52716	52778	52830	52841	52903	55865	55938	56000	56010	56073	59231	59295	59360	59369	59443
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	2.40	2.40	2.41	2.41	2.41	2.54	2.55	2.55	2.55	2.55	2.70	2.70	2.70	2.70	2.71	2.86	2.86	2.86	2.87	2.87
0°F (-17.8°C)	TC	51298	51367	51422	51424	51494	54389	54454	54507	54518	54583	57639	57713	57778	57789	57853	61111	61178	61244	61254	61330
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	2.44	2.44	2.44	2.44	2.45	2.58	2.59	2.59	2.59	2.59	2.74	2.74	2.74	2.74	2.75	2.90	2.91	2.91	2.91	2.91
5°F (-15.0°C)	TC	53271	53342	53400	53402	53474	56481	56548	56604	56615	56680	59855	59933	60000	60011	60078	63462	63531	63600	63610	63689
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	2.48	2.49	2.49	2.49	2.49	2.63	2.63	2.64	2.64	2.64	2.79	2.79	2.80	2.80	2.80	2.96	2.96	2.96	2.96	2.97
14°F (-10.0°C)	TC	55156	55231	55290	55293	55367	58480	58550	58608	58619	58689	61974	62055	62124	62136	62205	65708	65780	65851	65862	65944
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	2.80	2.81	2.81	2.81	2.81	2.97	2.97	2.98	2.98	2.98	3.15	3.15	3.16	3.16	3.16	3.34	3.34	3.35	3.35	3.35
17°F (-8.3°C)	TC	55785	55860	55920	55923	55998	59147	59217	59275	59287	59358	62681	62762	62832	62844	62914	66457	66529	66602	66612	66695
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	2.91	2.91	2.92	2.92	2.92	3.08	3.09	3.09	3.09	3.10	3.27	3.27	3.28	3.28	3.28	3.47	3.47	3.47	3.47	3.48
23°F (-5.0°C)	TC	55646	55721	55781	55783	55858	58999	59069	59127	59139	59209	62524	62605	62675	62687	62757	66291	66363	66436	66446	66529
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	2.92	2.93	2.93	2.93	2.93	3.10	3.10	3.11	3.11	3.11	3.28	3.29	3.29	3.29	3.30	3.48	3.49	3.49	3.49	3.49
35°F (1.7°C)	TC	55367	55441	55501	55504	55579	58704	58773	58831	58843	58913	62211	62292	62361	62373	62442	65959	66031	66103	66113	66195
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	2.95	2.95	2.96	2.96	2.96	3.13	3.13	3.13	3.14	3.14	3.32	3.32	3.32	3.32	3.33	3.51	3.52	3.52	3.52	3.53
41°F (5.0°C)	TC	54920	54994	55053	55056	55130	58230	58299	58356	58368	58437	61709	61789	61858	61869	61938	65427	65498	65569	65580	65661
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	3.14	3.14	3.15	3.15	3.15	3.33	3.33	3.33	3.34	3.34	3.53	3.53	3.53	3.53	3.54	3.74	3.74	3.75	3.75	3.75
47°F (8.3°C)	TC	54473	54547	54605	54608	54681	57756	57825	57882	57893	57962	61207	61286	61355	61366	61434	64895	64965	65036	65046	65127
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	3.33	3.33	3.33	3.33	3.34	3.53	3.53	3.53	3.53	3.54	3.74	3.74	3.75	3.75	3.75	3.96	3.97	3.97	3.97	3.98
50°F (10.0°C)	TC	53772	53844	53902	53905	53977	57013	57080	57137	57148	57216	60419	60497	60565	60576	60644	64059	64129	64199	64209	64289
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	3.33	3.33	3.33	3.33	3.34	3.53	3.53	3.53	3.53	3.54	3.74	3.74	3.75	3.75	3.75	3.96	3.97	3.97	3.97	3.98
59°F (15.0°C)	TC	51668	51738	51793	51796	51866	54782	54847	54901	54912	54977	58055	58130	58195	58206	58271	61553	61620	61687	61696	61773
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	3.32	3.33	3.33	3.33	3.34	3.52	3.53	3.53	3.53	3.54	3.73	3.74	3.74	3.74	3.75	3.96	3.96	3.97	3.97	3.97
68°F (20.0°C)	TC	49564	49631	49684	49687	49754	52551	52613	52665	52676	52738	55691	55763	55825	55836	55898	59046	59111	59175	59184	59258
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	3.32	3.33	3.33	3.33	3.34	3.52	3.53	3.53	3.53	3.54	3.73	3.74	3.74	3.74	3.75	3.96	3.96	3.97	3.97	3.97
77°F (25.0°C)	TC	47460	47524	47575	47578	47642	50321	50380	50430	50440	50500	53327	53396	53456	53466	53525	56540	56601	56663	56672	56742
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	3.32	3.33	3.33	3.33	3.33	3.52	3.53	3.53	3.53	3.53	3.73	3.74	3.74	3.74	3.75	3.96	3.96	3.97	3.97	3.97
82°F (27.8°C)	TC	46497	46560	46610	46612	46675	49299	49358	49406	49416	49475	52245	52313	52371	52381	52439	55392	55453	55513	55522	55591
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	3.50	3.51	3.51	3.51	3.52	3.71	3.72	3.72	3.72	3.73	3.94	3.94	3.95	3.95	3.95	4.17	4.18	4.18	4.18	4.19
86°F (30.0°C)	TC	45986	46048	46098	46100	46162	48757	48815	48863	48873	48931	51670	51738	51795	51805	51863	54784	54843	54903	54911	54980
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	3.88	3.88	3.89	3.89	3.89	4.11	4.12	4.12	4.12	4.13	4.36	4.36	4.37	4.37	4.37	4.62	4.63	4.63	4.63	4.64
95°F (35.0°C)	TC	44836	44896	44945	44947	45007	47538	47594	47642	47651	47707	50378	50444	50500	50509	50566	53414	53472	53530	53538	53605
	S/T	0.64	0.75	0.85	0.85	0.97	0.60	0.71	0.81	0.81	0.92	0.57	0.67	0.76	0.76	0.87	0.54	0.63	0.72	0.72	0.82
	PI	4.72	4.73	4.73	4.73	4.74	5.01	5.01	5.02	5.02	5.03	5.31	5.31	5.32	5.32	5.33	5.63	5.63	5.64	5.64	5.65
104°F (40.0°C)	TC	39232	39284	39327	39328	39382	41596	41645	41686	41695	41744	44081	44138	44187	44196	44245	46737	46788	46839	46846	46904
	S/T	0.69	0.81	0.92	0.92	1.00	0.65	0.76	0.87	0.87	0.99	0.61	0.72	0.82	0.82	0.93	0.58				

Climate 5000 Series - Gen 4

48K Btu Single Zone Regular Light Commercial System - 4-Way Cassette IDU



Heating Performance - Indoor Dry Bulb Temperature at 1235 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	25946	25427	25300	25199	24972	4.62	4.65	4.66	4.67	4.69
-4°F(-20.0°C)	32920	32261	32100	31972	31684	5.15	5.19	5.20	5.21	5.23
0°F(-17.8°C)	36748	36013	35833	35690	35369	5.33	5.38	5.39	5.40	5.42
5°F(-15.0°C)	41534	40704	40500	40338	39975	5.57	5.61	5.63	5.64	5.66
14°F(-10.0°C)	44688	43794	43575	43401	43010	5.97	6.02	6.03	6.04	6.07
17°F(-8.3°C)	45739	44824	44600	44422	44022	6.10	6.15	6.17	6.18	6.20
22°F(-5.6°C)	46541	45610	45382	45201	44794	5.99	6.04	6.05	6.07	6.09
27°F(-2.8°C)	47344	46397	46165	45980	45566	5.88	5.93	5.94	5.96	5.98
32°F(0.0°C)	48146	47183	46947	46759	46339	5.77	5.82	5.83	5.84	5.87
35°F(1.7°C)	48627	47655	47417	47227	46802	5.71	5.75	5.77	5.78	5.80
37°F(2.8°C)	51120	50098	49847	49648	49201	5.74	5.78	5.80	5.81	5.83
42°F(5.6°C)	57352	56205	55924	55700	55199	5.81	5.86	5.87	5.88	5.90
44.6°F(7.0°C)	59844	58647	58354	58121	57598	5.84	5.88	5.90	5.91	5.93
47°F(8.3°C)	63583	62312	62000	61752	61196	5.88	5.93	5.94	5.95	5.98
52°F(11.1°C)	61205	59981	59681	59442	58907	5.20	5.24	5.25	5.26	5.28
57°F(13.9°C)	58827	57650	57362	57133	56618	4.52	4.55	4.56	4.57	4.59
59°F(15.0°C)	57876	56718	56435	56209	55703	4.24	4.28	4.29	4.29	4.31
62°F(16.7°C)	56449	55320	55043	54823	54330	3.83	3.86	3.87	3.88	3.90
64.4°F(18.0°C)	55307	54201	53930	53714	53231	3.51	3.53	3.54	3.55	3.56
69.8°F(21.0°C)	52739	51684	51426	51220	50759	2.77	2.79	2.80	2.80	2.81
75.2°F(24.0°C)	50170	49167	48921	48725	48287	2.03	2.05	2.05	2.06	2.06

Bolded values were tested. All other values are simulated and for reference only.